

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Medium

Question

Rectangle A has length 15 and width w . Rectangle B has length 20 and the same length-to-width ratio as rectangle A. What is the width of rectangle B in terms of w ?

Answer

- A. $\frac{4}{3}w$
- B. $w + 5$
- C. $\frac{3}{4}w$
- D. $w - 5$

Correct Answer: A

Rationale

Choice A is correct. It’s given that rectangle A has length 15 and width w . Therefore, the length-to-width ratio of rectangle A is 15 to w . It’s also given that rectangle B has length 20 and the same length-to-width ratio as rectangle A. Let x represent the width of rectangle B. The proportion $\frac{15}{w} = \frac{20}{x}$ can be used to solve for x in terms of w . Multiplying both sides of this equation by x yields $\frac{15x}{w} = 20$, and then multiplying both sides of this equation by w yields $15x = 20w$. Dividing both sides of this equation by 15 yields $x = \frac{20w}{15}$. Simplifying this fraction yields $x = \frac{4}{3}w$.

Choices B and D are incorrect and may result from interpreting the difference in the lengths of rectangle A and rectangle B as equivalent to the difference in the widths of rectangle A and rectangle B. Choice C is incorrect and may result from using a length-to-width ratio of w to 15, instead of 15 to w .